هيئة تقويم التعليم والتدريب
Education & Training Evaluation Commission

2024
TPG-152

Annual Program Report

(Postgraduate Programs)

Program Name:	**MS in Cyber security & Digital Forensics**
Program Code (as per the Saudi Standard Classification of Educational Levels and Specializations):	**Level 7**
Qualification Level:	**M.Sc.**
Department:	**Information Technology**
College:	**College of Computer & Information Sciences**
Institution:	**Majmaah University**
Academic Year:	**2024-2025**
Main Location:	**College of Computer & Information Sciences**
Branches offering the Program (if any): N/A • …… …… • …… • ……	

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A. Program Statistics (in reporting year)

Item	Number
Number of students enrolled in the program	26 (Male:16, Female:10)
Number of students who started the program	26 (Male:16, Female:10)
Number of students who completed the program	10 (Male:03, Female:07)
Number of students who completed an intermediate award specified as an early exit point (if any)	N/A

*A separate cohort analysis report for male and female sections and for each branch (if any)

B. Program Assessment

1. Program Learning Outcomes Assessment and analysis according to PLOs assessment plan *

#	Program Learning Outcomes	Assessment Methods (Direct and Indirect)	Targeted Performance (%)	Assessment Results
Knowledge and Understanding				
K1	**An ability to understand a problem and identify the computing requirements appropriate to its solution**	Class Test, Mid Exam, Final Exam	**70%**	**89%**
Skills				
S1	Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions	Class Test, Mid Exam, Final Exam	**70%**	**89%**
S2	Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline	Lab Based Assignments, Mini Project	**70%**	**87%**
S3	Communicate effectively in a variety of professional	Group Assignments, Mini Project	**70%**	**77%**

	contexts			
S4	Identify and analyze user needs and to take them into account in the selection, creation, integration, evaluation, and administration of computing-based systems.	Case Study Implementation/ Laboratory /Mini project	**70%**	**80%**
S5	An ability to apply knowledge of computing and mathematics appropriate to the discipline	Class Test, Mid Exam, Final Exam	**70%**	**82%**
Values, autonomy, and responsibility				
V1	Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline	Oral or Written Communication, Seminar	**70%**	**92%**
V2	Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles	Class Test, Mid Exam, Final Exam	**70%**	**93%**
V3				
V..				

*Attach a separate report on the program learning outcomes assessment results for each semester (in the report year)، and for male and female sections and for each branch (if any).

Strengths:

- All PLOs meet or exceed the 80% target, with S3 (77%) and S4 (80%) as the lowest, prioritized for improvement
- Sustained excellence in Values (V1: 92%, V2: 93%) and research skills (S2: 87%).
- The 2024–25 cycle demonstrates robust PLO attainment (86–93% direct), exceeding the 80% target across most domains. Strengths in ethical values (V1, V2) and research skills (S2) align with strong KPI performance (e.g., 95% student publications).

Areas for improvement prioritized:

- **S3 (Problem-Solving)**: Lowest at 77%, driven by low performance in IT 603 (60%); needs course-level review.
- **S4 (Communication/Teamwork)**: At 80%, meets target but requires reinforcement through group projects.
- **K1 (Knowledge)**: Foundational courses (e.g., IT 601: 72%, IT 603: 70%) weaker than advanced courses.

2. Students' Evaluation of Courses

Course Code	Course Title	Number of Students Who Evaluated the Course	Percentage of Participants	Evaluation Results (out of 10)	Developmental Recommendations
IT 601	Cybersecurity Foundations	25	100%	8.7	No specific recommendation
IT 602	Digital Forensics Technology and Practices	25	100%	8.9	No specific recommendation
IT 603	Information Security Law, Policy, and Ethics	25	100%	8	No specific recommendation
IT 604	Cybersecurity Technology and Management	12	100%	9	Fous on latest case studies on GenAI challenges and opportunities in security
IT 605	Research Methods and Engineering Statistics	12	100%	9	
IT 611	Network and System Security	11	100%	8.9	No specific recommendation
IT 624	Malware Development and Countermeasures	11	100 %	9	No specific recommendation

Course Code	Course Title	Number of Students Who Evaluated the Course	Percentage of Participants	Evaluation Results (out of 10)	Developmental Recommendations
IT 616	Special Topics in Cybersecurity	10	100%	9	No specific recommendation
IT 621	Computer Forensics and Investigations	10	100%	9.1	No specific recommendation
IT 650	Research Project	10	100%	9.4	No specific recommendation

3. Students Evaluation of Program Quality

Evaluation Date:	Number of Participants:
Students Feedback	**Program Response**
Strengths: • Students are satisfied with the teaching and learning process. • Labs are available to all students • Every course is assigned differ faculty member • Course Quality (PG-2): Average 4, approaching 4.5 and maintaining a solid base. • Student Services (PG-7): 71% satisfaction, approaching 90% with targeted enhancements.	**14**
Areas of Improvement: • Course Quality Evaluation • Academic Supervision • Student Services Satisfaction	
Suggestions for improvement: • **Teaching & Learning**: Revise course content and supervision methods, Faculty development workshops • **Student Support & Retention:** Address dropout causes via targeted interventions, improve student services, especially for female students	

• **Research & Innovation:** Incentivize publications and student-faculty collaboration, Leverage recognition awards to enhance visibility •	

4. Academic Research and innovation during the reporting year

A .Research Achievements (in the year concerned)

Activities Implemented	Number
Published scientific research	125
Research projects	10 student research projects
Conferences organized by the program	-
Seminars held by the program	-
Conferences attendees	-
Seminars attendees	-

* Attach a research production statement of the faculty and students in the program including basic data such as (researcher's name, research title, publishing entity, publishing date, etc.)

** In the case of multiple scientific production, the program lists the numbers for each classification individually (Example in individual research: if the total number is (10) in the case of production (6) published, and (4) acceptable for publication, etc.)

- Discussion and analysis of scientific research and innovation activities

Positive Trends

1. **KPI-PG-10 (Published Research per Faculty Member):**
   Increased from **9 (2023–24)** to **10 (2024–25)**, showing strong faculty engagement in research.

2. **KPI-PG-11 (Citations per Faculty Member):**
   Improved from **2.5 to 3.0**, indicating growing research visibility and impact.

3. **KPI-PG-12 (Students' Publications):**
   Increased significantly from **79% (2023–24)** to **~95% (2024–25, under review)**, reflecting enhanced student participation in research dissemination.

4. **KPI-PG-13 (Patents/Innovations/Awards):**
   Grew from **1 (2023–24)** to **3 (2024–25)**, including recognition of faculty in the *Top 10% Scientist Award*

B .Theses (approved during the year concerned) N.A.

Thesis Title	Researcher's Name	Supervisor's Name	Grade
A Blockchain-Driven NLP and ML Framework for Adaptive and Explainable Cyber Threat Intelligence Systems	Ruba Ahmed Alfahidah	Dr.Shailendra Mishra	A+
Explainable Deep Learning with CNN-LSTM and LIME for Securing IIoT Ecosystems	Ebtesam Abdulaziz Almutairi	Dr.Shailendra Mishra	A+
Securing Digital Interactions: LSTM-Driven Classification of Deepfake Voice Audio	Reham	Dr.A.HaQ	A+
Mitigating AI-Driven Cybersecurity Vulnerabilities Exploited by LLM Agents	Raghad	Dr.A.HaQ	A+
Secure Smart Meters Based on Threat Modelling	Dalal Aldablan	Dr.Shakeel	A+
Deepfake Image Detection Using Advanced Convolutional Neural Networks: Addressing Emerging Cybersecurity Threats	Alya	Dr.A.HaQ	A+
Real-Time Cybersecurity Threat Prediction Using NLP and Deep Learning	Nouf	Dr.A.Haq	B

Thesis Title	Researcher's Name	Supervisor's Name	Grade
Reliable Detection of IoT Cyberattacks via Packet Flow Analysis	Nahed	Dr.A.Haq	A
ICRDS: A Novel Intelligent Cyber Risk Detection System for Proactive Threat Mitigation	Omar Ahmad Alyousef	Dr.Shailendra Mishra	A
Comprehensive Analysis of Cybersecurity Attack Trends: Uncovering Patterns for Proactive Defense	Saif	Dr.A.Haq	A

- Comment on Theses

The master's research project outputs demonstrate encouraging progress across supervisors, with several papers already submitted to reputable journals.

Under the supervision of Dr. Shailendra Mishra, three projects have reached advanced stages: *Ruba Ahmed Alfahidah's* work on a Blockchain-Driven NLP and ML Framework, *Ebtesam Abdulaziz Almutairi's* research on Explainable Deep Learning with CNN-LSTM and LIME for IIoT security, and *Omar Ahmad Alyousef's* Intelligent Cyber Risk Detection System (ICRDS), all submitted to *PeerJ Computer Science* and currently under review with initial feedback already addressed. These projects reflect strong novelty and integration of blockchain, explainable AI, and proactive threat detection.

Similarly, under Dr. A. Haq, several projects address pressing issues in AI-driven security threats, including deepfake voice and image detection, LLM-based vulnerabilities, IoT attack detection, and trend analysis. While some papers such as those by *Reham*, *Raghad*, and *Alya* are under review, others by *Nouf*, *Nahed*, and *Saif* received rejection with recommendations for improvement, primarily due to limited novelty and insufficient dataset validation. Strengthening experimental rigor and incorporating diverse datasets such as CICIDS 2017, UNSW-NB15, or real-world data will enhance their resubmission prospects.

Meanwhile, under Dr. Shakeel, *Dalal Aldablan's* project on Secure Smart Meters based on Threat Modelling has been submitted to *IJCSE (Inderscience)* and is awaiting review. Overall, the portfolio highlights timely topics—blockchain security, explainable AI, IoT, and deepfake detection—with several promising publications in the pipeline. The primary strength lies in problem selection and

alignment with global cybersecurity challenges, while improvement is required in methodological depth, dataset justification, and clearer articulation of novelty to ensure stronger acceptance rates.

5. Students' Evaluation of Academic Supervision

Evaluation Date:	Number of Participants:
Students Feedback	**Program Response**
Strengths: • The overall average (4.2/5) reflects a *very good* level of satisfaction with academic supervision across the postgraduate program. • Female students reported slightly higher satisfaction (4.4) compared to male students (4.0). This positive outcome suggests that supervisors are effectively guiding research students; however, additional efforts may be directed towards addressing the concerns of male students to further balance and enhance the supervision experience.	**10**
Areas of Improvement: • **Individualized Support:** Some male students rated supervision slightly lower (4.0 vs. 4.4). Supervisors may increase one-to-one mentoring sessions, focusing on research challenges and career guidance. • **Timely Feedback:** Strengthen the consistency and timeliness of supervisors' feedback on drafts, progress reports, and project milestones. • **Research Methodology Guidance:** Provide additional structured workshops on research design, data analysis, and academic writing to complement supervision. • **Supervisor–Student Communication:** Encourage supervisors to schedule regular meetings with clear agendas to ensure alignment and track progress more effectively. • **Monitoring & Review:** Implement mid-semester review surveys for early identification of supervision gaps and immediate corrective actions.	
Suggestions for improvement :Establish a Supervision Quality Enhancement Plan, including training sessions for supervisors on mentoring best practices, cross-gender	

supervision balance, and standardized supervision guidelines, to further elevate student satisfaction in future cycles	

6. Other Evaluation (if any)-NIL

(e.g., independent reviewer, program advisory committee, and stakeholders (e.g., faculty members, alumni, and employers)

Evaluation method:	Date:	Number of Participants:

Summary of Evaluator Review	Program Response
Strengths: • •	
Points for Improvements: • •	
Suggestions for development: • •	

*Attach independent reviewer's report and stakeholders' survey reports (if any).

C. Program Key Performance Indicators (KPIs)

Including the key performance indicators required by the NCAAA.

Master of Cybersecurity and Digital Forensics – Program KPIs (2024-25)

Standard	Code	Key Performance Indicators	Description	Actual performance	Targeted performance level	Internal reference (Internal benchmark (Previous years 2023-24).	External reference (External benchmark 2022-23*)	New target performance level
2 - Teaching and	**KPI - PG-1**	**Students' Evaluati**	Average of overall rating of final year students	**M-4.0** **F-4.4** **AV-4.2**	4.5	M-4.4 F-4.1 **AV-4.3**	4.45	4.5

Learning		**on of quality of learning experience in the program**	for the quality of learning experience in the program on a five-point scale in an annual survey					
	KPI-PG-2	**Students' evaluation of the quality of the courses**	Average students overall rating for the quality of courses on a five-point scale in an annual survey	**M-4.1** **F-3.8** **AV-4**	4.5	M-4.5 F-4.2 **AV-4.35**	4.34	4.5
	KPI-PG-3	**Students' evaluation of the quality of academic supervision**	Average students' overall rating of the quality of scientific supervision on a five-point scale in an annual survey.	**M-4.0** **F-4.4** **AV-4.2**	80%	M-4.5 F-4.4 **AV-88%**	94%	80%
	KPI-PG-4	**Average time for stud**	Average time (in seme	**2 years**	2 ye	2 years	2 Years	2 ye

		ents' graduation	sters) spent by students to graduate from the program.		ars			ars
	KPI-PG-5	**Rate of students dropping outof the program (Data Needed)**	The percentage of students who did not complete the program of the total number of students in the same cohort.	16%	10%	M-20% F-0% **AV-10%**	5%	10%
	KPI-PG-6	**Employers' evaluation of the program graduates' competency**	Average of overall ratingof employers for the competency of the program graduates on a five point scale in an annual survey.	4.6	4	4.6	Not available	4
-3-Students	**KPI-PG-7**	**Students' satisfaction with the provi**	Average of students' satisfaction rate with the various	**M-83% F-62% AV-71.42%**	90%	M-92% F-85% **AV-88%**	89.6%	90%

		ded servic es	services provided by the program (food, transpor tation, sport facilities, academi c advising,) an annual survey.					
-4- FAC ULT Y ME MB ERS	**K PI - P G- 8**	**Ra ti o of st ud en ts to fa cu lty m e m be rs**	Ratio of the total number of students to the total number of full-time and full-time equivalen t faculty members participati ng in the program.	**M-3:1** **F-15:1** **AV-5.1**	6 : 1	M-3:1 F-15:1 **AV-5.1**	3:1	6 : 1
-6- RESEA RCH AND PROJE CTS	**K PI - P G- 9**	**Perc enta ge of publi catio ns of facul ty mem bers**	Percenta ge of faculty members participa ting in the program with at leastone research publicati on during the year	**M-90%** **F-100%** **AV-90%**	6 0 %	M-89% F-100% **AV-90%**	79. 17 %	6 0 %

			to total faculty members in the program.					
	KPI-PG-10	**Rate of published research per faculty member**	The average number of refereed and/or published research per each faculty member participating in the program during the year (total number of refereed and/or published research to the total number of faculty members during theyear)	**M-10** **F-6** **AV-10**	1	M-89% F-100% **AV-9**	2.33	2
	KPI-P	**Citations rate in refereed**	The average number of citations in refereed journals	**M-3.2** **F-1.2** **AV-3**	1	M-2.7 F-0.2 **AV-2.5**	1.07	3.5

	G-11	**journals per faculty member**	from published research (total number of citations in refereed journals from published research for faculty members to the total published research (					
	KPI-PG-12*	**Percentage of students' publication**	Percentage of students who: a. published their	M-95% UNDER PROGRESS (Paper	100%	M-92% F-57% **AV-79%**	Not Available	50%

			r rese arch in refe ree d jo ur n al s b. pre s e n t e d p a p e r s i n c o n f e r e n c e s to the total number of students in the program during the year.	s a r e u n d e r r e v i e w s t a g e) F - 1 0 0 % U N D E R P R O G R E S S (P a p e r s a				

				r e u n d e r r e v i e w s t a g e)				
	K PI - P G - 1 3	**Number of patents, innovative products, and awards of excellence**	Number of: a. Patents and innovative products National and international excellence awards obtained annually by the students and staff of the program.	**3****	1	-	**1**	**1**

**-Top 10% Scientist Award (3 Faculty Members awarded Top 10% Scientist Award-Stanford University Database-Facilitated by Rector Majmaah University)

* The King Abdulaziz University's MS Cybersecurity program is chosen as an external benchmark as it is close to our program.

- Comments on the Program KPIs and Benchmarks results:

Trend Analysis – Key Observations

Positive Trends

1. **KPI-PG-10 (Published Research per Faculty Member):**
   Increased from **9 (2023–24)** to **10 (2024–25)**, showing strong faculty engagement in research.
2. **KPI-PG-11 (Citations per Faculty Member):**
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3. **KPI-PG-12 (Students' Publications):**
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4. **KPI-PG-13 (Patents/Innovations/Awards):**
   Grew from **1 (2023–24)** to **3 (2024–25)**, including recognition of faculty in the *Top 10% Scientist Award*.

Stable Trends

1. **KPI-PG-4 (Graduation Time):**
   Remains consistent at **2 years**, aligned with program expectations.
2. **KPI-PG-6 (Employers' Evaluation of Graduates' Competence):**
   Stable at **4.6**, above target, confirming strong employer satisfaction.
3. **KPI-PG-9 (Faculty Research Publications):**
   Maintained at **90%**, consistently exceeding the target of 60%.
4. **KPI-PG-8 (Student-to-Faculty Ratio):**
   Remains at **5.1:1**, well within acceptable thresholds.

Negative Trends

1. **KPI-PG-1 (Learning Experience Evaluation):**
   Slight decline from **4.3 to 4.2**, indicating the need to strengthen student engagement and teaching quality.
2. **KPI-PG-2 (Course Quality Evaluation):**
   Decreased from **4.35 to 4.0**, highlighting concerns in course design and delivery.
3. **KPI-PG-3 (Academic Supervision Quality):**
   Dropped from **88% to 4.2 (scale-based)**, requiring improved academic advising and mentoring.
4. **KPI-PG-5 (Student Dropout Rate):**
   Increased from **10% to 16%**, signaling a major area of concern regarding retention and student support.

5. **KPI-PG-7 (Student Services Satisfaction):**
   Declined sharply from **88% to 71.4%**, particularly due to lower satisfaction among female students.

D. Challenges and difficulties encountered by the program (if any)

Teaching	**Keeping Up with Rapid Technological Changes Balancing Theory and Practice, Diverse Student Backgrounds**
Assessment	**Designing Effective Assessment Methods**
Guidance and counseling	**Limited Faculty Availability**
Learning Resources	**Access to Up-to-Date Resources**
faculty	**Continuous Professional Development**
Research Activities/thesis supervision	**Identifying Relevant Research Topics**
Others	

E. Program development Plan

1. Teaching & Learning

KPI	Issue	Action	Responsibility	Timeline	Success Metric
PG-1, PG-2	Decline in learning experience (4.3→4.2) and course quality (4.35→4.0)	Revise course content; integrate student feedback; faculty pedagogy training workshops	HoD IT, Program Coordinator, Quality Committee	6–12 months	≥4.4 average in student surveys
PG-3	Drop in academic supervision quality (88%→4.2/5)	Introduce structured supervision framework, regular advising sessions, faculty mentoring guidelines	HoD IT, Program Coordinator	6 months	≥85% student satisfaction in supervision

2. Student Retention & Services

KPI	Issue	Action	Responsibility	Timeline	Success Metric
PG-5	Dropout rate increased (10%→16%)	Conduct exit interviews; provide academic counseling; introduce flexible learning support	Student Affairs, Program Coordinator	Ongoing (Review each semester)	Dropout ≤ 8%
PG-7	Decline in student services satisfaction (88%→71.4%)	Enhance facilities, improve responsiveness to female student needs, introduce digital support services	HoD IT, Student Services Unit	1 year	≥85% satisfaction in annual survey

3. Research & Innovation

KPI	Issue	Action	Responsibility	Timeline	Success Metric
PG-9–PG-11	Research output improved, but sustainability required	Maintain research incentives, collaborative projects, journal publication workshops	Research Committee, Faculty	Ongoing	≥90% faculty with publications; citations ≥3.5
PG-12	Student research under review (95%)	Support journal submission, track progress until acceptance, mentorship by faculty	Supervisors, Research Unit	12 months	≥90% students with accepted publications
PG-13	Patents/awards increased (1→3)	Expand innovation support fund, encourage patent filing workshops	Quality Commitee, Program Faculty	1–2 years	≥4 annual recognitions

4. Employer Engagement

KPI	Issue	Action	Responsibility	Timeline	Success Metric
PG-6	Stable at 4.6 but must sustain	Conduct employer roundtables, review graduate skill alignment, embed industry projects	Dean, HoD IT	1 year	≥4.5 employer rating maintained

*Attach any unachieved improvement plans from previous report.
** The annual program report should to be discussed and approved by department council

F. Approval of Annual Program Report

COUNCIL / COMMITTEE	IT DEPARTMENT COUNCIL
REFERENCE NO.	**5TH MEETING**
DATE:	**22-09-2025**